Week 7 HW Question 10: Date:

Given: $u = (4, 11)$, $v = (5, 3)$, $\langle u, v \rangle = u \cdot v$

a.) Does the Cauchy-Schwarz inequality hold? $|u \cdot v| \leq \|u\| \, \|v\|$

$u \cdot v = (4)(5) + (11)(3) = 20 + 33 = 53$

$\|u\| = \sqrt{4^2 + 11^2} = \sqrt{137}$, $\|v\| = \sqrt{5^2 + 3^2} = \sqrt{34}$

$\|u\| \, \|v\| = \sqrt{137} \cdot \sqrt{34} \approx 68.25$

$|53| \leq 68.25$ so, yes, the Cauchy-Schwarz inequality holds.

b.) Does the triangle inequality hold? $\|u + v\| \leq \|u\| + \|v\|$

$u + v = (9, 14)$, $\|u + v\| = \sqrt{9^2 + 14^2} = \sqrt{277} \approx 16.64$

$\|u\| = \sqrt{4^2 + 11^2} = \sqrt{137}$, $\|v\| = \sqrt{5^2 + 3^2} = \sqrt{34}$

$\|u\| + \|v\| = \sqrt{137} + \sqrt{34} \approx 11.70 + 5.83 = 17.53$

$16.64 \leq 17.53$ so, yes, the triangle inequality holds.